FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA TEST SERIES

FIITJEE

JEE (Advanced), 2015

CONCEPT RECAPITULATION TEST - IV

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 180**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 – 10)** contains 10 multiple choice questions which have one or more than one correct answer. Each question carries **+3 marks** for correct answer. There is no negative marking.

(ii) **Section-C (01 – 10)** contains 10 Numerical based questions with answers as numerical value from **0 to 9** and each question carries **+3 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

(One or More than One Options Correct Type)

This section contains **10 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE THAN ONE** are correct.

1. A particle is released at x = 1 in a force field $\vec{F}(x)=\left(\frac{2}{x^2}-\frac{x^2}{2}\right)\hat{x}$, $x \geq 0$. Which of the following statements is/are true?
   (A) $\vec{F}(x)$ is conservative
   (B) Angular momentum of the particle about the origin is const.
   (C) The particle moves towards $x=\sqrt{2}$
   (D) The particle moves towards the origin

2. Light from a monochromatic source is incident normally on a small photo sensitive surface S having work function ϕ. If power of the source is W and a is the distance between the source and S, then
   (A) the number of photons striking the surface per unit time will be $\left(\frac{w\lambda S}{4\pi hca^2}\right)$
   (B) the maximum energy of the emitted electrons will be $\frac{1}{\lambda}(hc-\lambda\phi)$
   (C) the stopping potential needed to stop the most energetic photons will be $\frac{1}{e\lambda}(hc-\lambda\phi)$
   (D) photo emission occurs only if $0 \leq \lambda \leq \frac{hc}{\phi}$

3. In a one-dimensional collision between two identical particles A and B, B is stationary and A has momentum P before impact. During impact, B gives impulse J to A.
   (A) The total momentum of 'A plus B' system is P before and after the impact and (P – J) during the impact
   (B) During the impact, A gives impulse J to B
   (C) The coefficient of restitution is $\frac{2J}{P}-1$
   (D) The coefficient of restitution is $\frac{J}{P}-1$

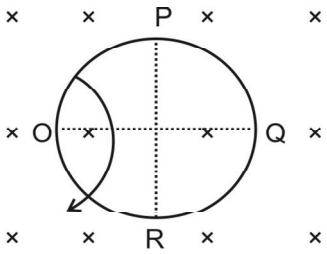
Space for Rough work

4. The electric potential decreases uniformly from 120V to 80V as one moves from y-axis from y = –2 cm to y = +2 cm. Then at origin

(A) Electric field must be equal $20\frac{V}{cm}$ (B) Electric field may be equal to $20\frac{V}{cm}$

(C) Electric potential must be 100V (D) Electric potential may be 60V

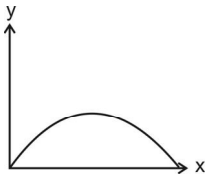
5. A conducting ring of radius 'a' is rotated about a point O on its periphery as shown in the figure in a plane perpendicular to uniform magnetic field $\vec{B}$ which exists everywhere. The rotation velocity is w, then:



(A) $v_P - v_0 > 0$ and $v_R - v_0 < 0$ (B) $v_p = v_R > v_0$

(C) $v_0 > v_p = v_Q$ (D) $v_Q - v_p = v_p - v_0$

6. A reflecting surface is represented by equation $Y = \frac{2L}{\pi}\sin\left(\frac{\pi x}{L}\right)$. A ray travelling horizontally becomes vertical after refection. The coordinates of the point(s). Where this ray is incident :

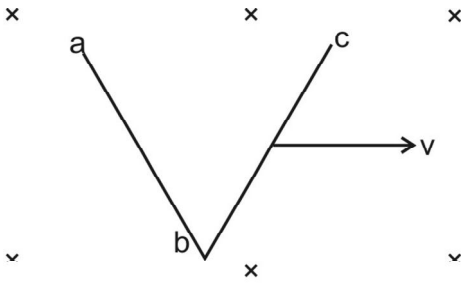


(A) $\left(\frac{L}{4},\frac{\sqrt{2}L}{\pi}\right)$ (B) $\left(\frac{L}{3},\frac{\sqrt{3}L}{\pi}\right)$

(C) $\left(\frac{3L}{4},\frac{\sqrt{2}L}{\pi}\right)$ (D) $\frac{2L}{3},\frac{\sqrt{3}L}{\pi}$

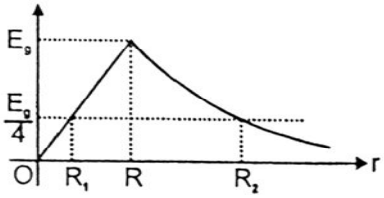
7. A V-shaped conducting wire is moved inside a magnetic field as shown in figure. Magnetic field is perpendicular to paper inwards. Then :

(A) $V_a = V_c$
(B) $V_a > V_c$
(C) $V_a > V_b$
(D) $V_c > V_b$



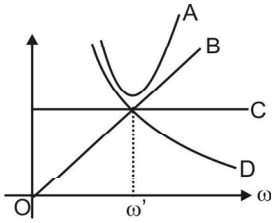
Space for Rough work

8. The variation of gravitational field intensity with distance from the centre of a body is shown in the graph from which one can conclude that

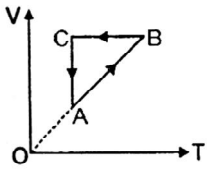


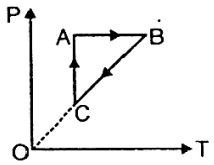
(A) Variation of gravitational field intensity is due to the spherical mass body of radius R
(B) $E_g \propto r$ for $r < R$
(C) the separation of two points having field $\frac{E_g}{4}$ may be $\frac{9R}{4}$
(D) the separation of two points R_1 is R is R/4

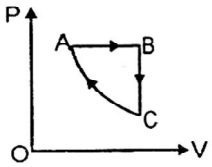
9. For an LCR circuit
(A) A and B represent R and Z respectively
(B) A and B represent Z and R respectively
(C) A, B, C and D represent Z, X_L, R and X_c respectively
(D) for $\omega = \omega'$, the phase difference between current and voltage becomes zero

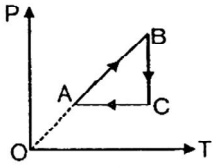


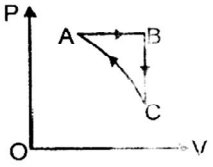
10. An ideal gas undergoes at thermodynamics cycle as shown in figure. Which of the following graphs represents the same cycle?



(A)


(B)


(C)


(D)


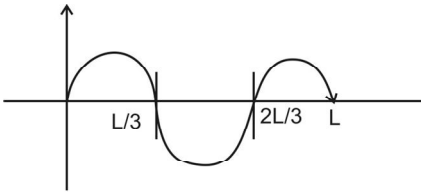
Space for Rough work

SECTION – C
(One Integer Value Correct Type)

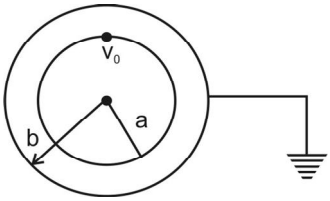
This section contains **10 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. A travelling pulse is given by $f(x, t) = A \exp\left(\frac{2abxt - a^2x^2 - b^2t^2}{c^2}\right)$, Where A, a, b, c are positive constants of appropriate dimensions, the speed of the pulse is nb/2a, n =

2. A free particle of mass 'm' is confined to a region of length L. The de-Broglie wave associated with the particle is sinusoidal in nature as given in figure the energy of the particle is $\frac{n^2h^2}{8mL^2}$, n =

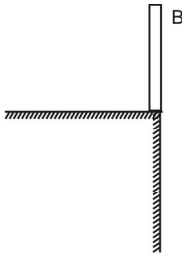


3. A concentric spherical volume of inner radius a and outer radius 'b' is filled with a material of finite conductivity specified by $\sigma(r) = \frac{A}{r^2}$ where A is a positive constant of appropriate dimensions.



The outer surface is grounded and the inner surface is maintained at potential V_0. The resistance of this configuration is $(b–a)/n\pi A$, n = _____.

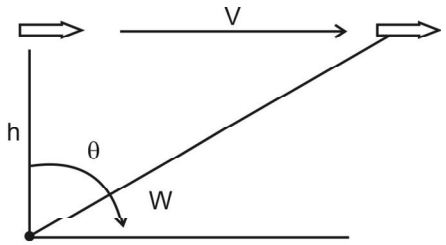
4. A rod of length 5m is placed at the edge of a smooth table as shown in figure. It is hit horizontally at point B. If the displacement of centre of mass in 1S is $5\sqrt{2}$ m, then the angular velocity of the rod is 5n rad/s, n = : (Given : g = 10 m/s^2)



5. Two consequent plane waves of light of equal amplitude and each of wavelength $20\pi\times10^{-8}$m propagating at an angle of $\frac{\pi}{1080}$ rad with respect to each other, fall almost normally on screen the fringe width (in mm) on the screen is 108/100n, n = _____.

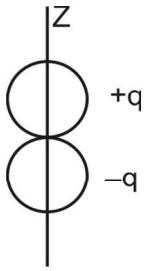
Space for Rough work

6. A directed beam from a small but powerful searchlight placed on the ground tracks a small plane flying horizontally at a fixed height 'h' above the ground with a uniform velocity V as shown in figure below.

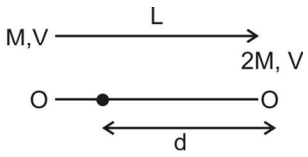


If the searchlight starts rotating with an instantaneous velocity ω_0 at t = 0, when the plane was directly overhead, then at a large time t, instantaneous angular velocity w(f) is given by $n\frac{w_0}{1+w_0^2t^2}$, n = _____

7. An electric dipole is constructed by fixing two circular charged rings, each of radius a, with an insulating contact, one of these rings has a total charge +Q and the other has total charge –Q. If the charge is distributed uniformly along each ring, the dipole moment about the point of contact will be $nQa\hat{Z}/4$, n = _____.



8. A dumbbell is placed in water of density ρ. It is observed that by attaching a mass m to the rod, the dumbbell floats with rod horizontal on the surface of water and each sphere exactly half submerged as shown in figure.



The volume of the mass m is negligible the value of length L is $nd\frac{(\rho V-3M)}{(\rho V-2M)}$, n = _____.

9. A small sphere of mass m and radius r rolls without slipping on the inside a large hemisphere of radius R. The axis of symmetry of the hemisphere is vertical. It starts at the top from rest. The small sphere will exert a normal force on the hemisphere at its bottom equal to $n\frac{17}{14}mg$, n = _____.

10. An electron of mass 'm' and charge 'e' initially at rest gets accelerated by a constant electric field E. The rate of change of De Broglie wavelength of this electron at time 't' is $\left(-\frac{h}{eE}t^{-n}\right)$; where n equals to_____.

Space for Rough work

Chemistry

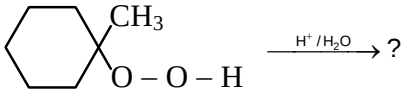
PART – II

SECTION – A

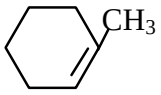
(One or More than One Options Correct Type)

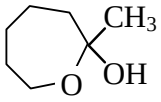
This section contains **10 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE THAN ONE** are correct.

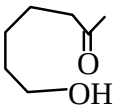
1. The product of the following reaction,

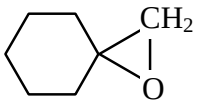


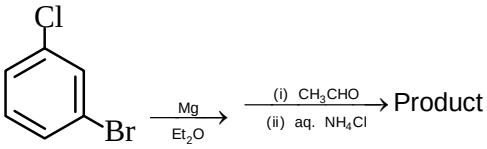
would be

(A) 

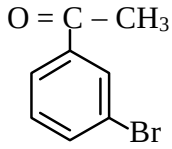
(B) 

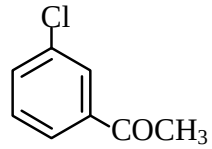
(C) 

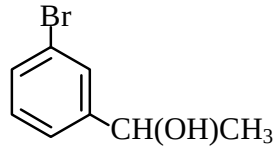
(D) 

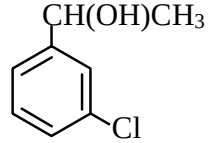
2. 

The product of above reaction is

(A) 

(B) 

(C) 

(D) 

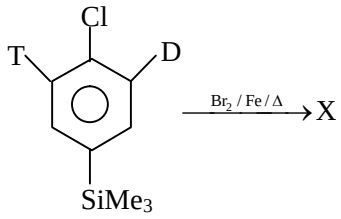
3. When 150 ml of ozonized oxygen was passed through red hot tube, the volume increased by 10 ml then the volume percentage of ozone in the sample is

(A) 25% (B) 51%

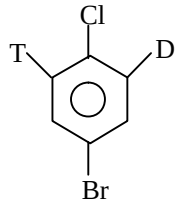
(C) 13.33% (D) 17.84%

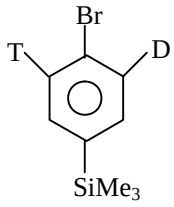
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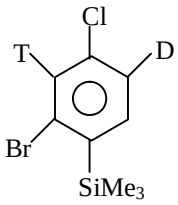
4. The volume of water needed to dissolve 1 mg of $PbSO_4$ ($K_{sp} = 1.44 \times 10^{-8}\ M^2$) at 25°C is approximately (molar mass of Pb = 207)
(A) 10 ml (B) 27 ml
(C) 43 ml (D) 80 ml

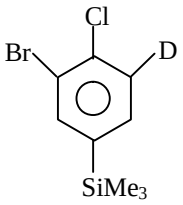
5. 

The product of the reaction (X) will be

(A) 

(B) 

(C) 

(D) 

6. Which of the following is a salt of sulphurous acid?
(A) $NaHSO_4$ (B) $Na_2S_2O_3$
(C) $Na_2S_4O_6$ (D) Na_2SO_3

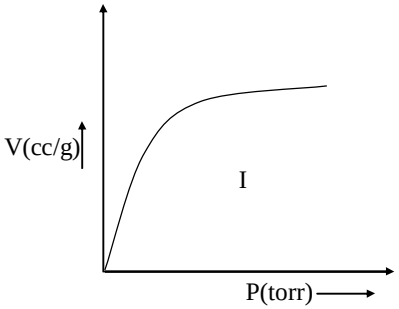
7. Which of the following is/are true for 3–Methyl butanone
(A) This compound is an isomer of 4–penten–1–ol
(B) It may be prepared by acidic Hg^{2+} catalyzed hydration of 3–methyl–1–butyne
(C) It may prepared by CrO_3 oxidation of 2–methyl–2–butanol
(D) It can not give positive iodoform test.

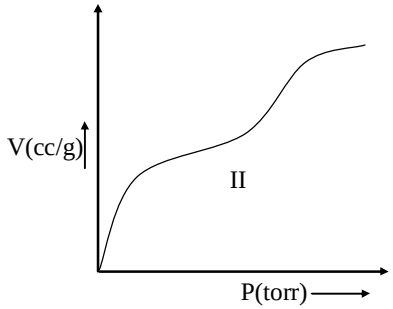
8. In blast furnace a charge of oxide ore, limestone and coke is feed for metallurgical extraction of iron. The correct statements regarding the process is/are
(A) at the top of furnace the reducing agent is CO
(B) at the top of furnace the reducing agent is C
(C) at the top of furnace the reducing agent is CO_2
(D) at the bottom of furnace the reducing agent is C

Space for Rough work

9. The correct facts regarding shape and hybridization of complexes $[CuCl_5]^{3-}$ and $[Ni(CN)_5]^{3-}$ will be
(A) both have similarly hybridized states on central metal ion
(B) both have different shapes
(C) both have different hybridized states on central metal ion
(D) both have similar shapes

10. On the basis of following graphs between volumes of gas adsorbed and pressure of gas





which of the following is/are correct ?
(A) graph I is of chemisorption
(B) graph II is of physiosorption
(C) graph II is of chemisorption
(D) graph I is of physiosorption

Space for Rough work

SECTION – C
(One Integer Value Correct Type)

This section contains **10 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. $A \xrightarrow[H_2SO_4]{NaCl} B\uparrow \xrightarrow{NaOH}$ yellow solution $\xrightarrow[2.\ H_2O_2]{1.\ H^+}$ Blue solution.
(red vapours)
The blue colour obtained at the end of reaction is due to salt formation. The oxidation number of the metal in the salt is

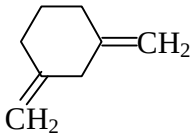
2. Find the quantum numbers of the excited state of electrons in He^+ ion which on transition to ground state and first excited state emit two photons of wavelengths, 30.4 nm and 108.5 nm respectively. ($R_H = 1.09678 \times 10^7\ m^{-1}$)

3. Electron is now a day considered as probability wave function with quantized angular momentum and energy. The angular momentum of e^- in 1s orbital is.

4. The number of atomic orbitals involved in the hybridization at central metal ion of $K_3[Fe(CN)_6]$ are

5. In an acidic buffer the ratio [acid]/[salt] is 0.5. If the concentration of acid make twice of its initial value then what is value of $10\Delta pH$. [log 2 = 0.3]

Space for Rough work

6. How many stereoisomers are obtained on complete hydrogenation of  ?

7. 20 ml solution containing mixture of NaOH and Na_2CO_3 require 60 ml of 0.5 N H_2SO_4 for the end point using phenolphthalein as indicator. After this when same solution was titrated using methyl orange indicator. A further 20 ml of 0.5 N H_2SO_4 was needed for the second end point. Calculate the molarity of NaOH.

8. A solution containing 2.68×10^{-3}mol of A^{n+}ions requires 1.61×10^{-3}mol of MnO_4^- for the complete oxidation of A^{n+} to AO_3^- in acidic medium. What is the value of n?

9. 0.15 mole of pyridinium chloride has been added into $500cm^3$ of 0.2M pyridine solution. What is the pH of the resulting solution assuming no change in volume. (K_b for pyridine $= 1.5\times10^{-9}$)

10. A solution of 6.2 g ethylene glycol in 55 g H_2O is cooled to -3.72^0C. What is the weight (in grams) of the ice separated from the solution?
($K_f H_2O = 1.86 k\,mole^{-1}kg$)

Space for Rough work

Mathematics

PART – III

SECTION – A

(One or More than One Options Correct Type)

This section contains **10 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE THAN ONE** are correct.

1. If curve $A \equiv \{z : |z-3|+|z+3| = 8\}$, where $B \equiv \{z : |z-3| = k\}$, $k \in R^+$ touches A internally and $C \equiv \{z : ||z-3|-|z+3|| = 4\}$, then
   (A) B also touches C internally
   (B) A and C intersect orthogonally
   (C) $k \in \{1\}$
   (D) $1 < k < 2$

2. If f(x) is a polynomial of degree three with leading coefficient as unity such that f(1) = 1; f(3) = 9 and f'(1) = 2, then
   (A) f′(x) = 2x has at least one root between (1, 3)
   (B) f″(x) = 2 has at least one root between (1, 7/3)
   (C) product of roots of f(x) = 0 is 4
   (D) sum of root of f(x) = 0 is 3

3. The inequality $\sin|z| > 0$ may represent, where z is complex number
   (A) an interior region of circle where centre is origin and whose radius is π
   (B) a parabola whose vertex in (0, 0)
   (C) a region between two concentric circles cantered at (0, 0) and having radii 2π and 3π
   (D) an ellipse of semi axes are π and 2π

4. A straight line touches the rectangular hyperbola $9x^2 - 9y^2 = 8$ and parabola $y^2 = 32x$ then equation of line is (are)
   (A) $9x + 3y - 8 = 0$
   (B) $9x - 3y + 8 = 0$
   (C) $9x + 3y + 8 = 0$
   (D) $9x - 3y - 8 = 0$

5. Which of the following is true about matrix?
   (A) if A is a non–singular square matrix of order $n \times n$ and k is scalar $(k, n) \neq (0, 1)$ then $|\text{adj}(kA)|$ is equal to $k^{n^2-n} |A|^{n-1}$
   (B) let A and B be 3×3 order matrices. If $|A| = 0$, $AB = O$ and $AC = O$ then $B = C$
   (C) A and B are two square matrixes of order $n \times n$ such that $AB = O$ and A is non-singular then B must be null matrix
   (D) if AB is non–singular and A, B are square matrixes. Then A and B both must be non–singular matrices

Space for rough work

6 For a given parabola $y^2 = 4ax$, two variable chords PQ and RS at right angles are drawn through the fixed point $A(x_1, y_1)$ inside the parabola, making variable angles θ and α with x-axis. If r_1, r_2, r_3, r_4 are distances of P, Q, R and S from A, then the value of $\frac{1}{r_1 r_2} + \frac{1}{r_3 r_4}$

(A) independent of θ (B) independent of α
(C) depends upon both θ and α (D) is a constant

7. Consider the equation $\tan(5\pi \cos\theta) = \cot(5\pi \sin\theta)$ for $\theta \in (0, 2\pi)$, then
(A) number of real solution is 28 (B) number of real solution is 14
(C) $\sin\theta + \cos\theta = \frac{2n+1}{10}$ (D) $\sin\theta - \cos\theta = \frac{2n+1}{10}$

8. A is non singular square matrix of $(n \times n)$ such that $A^2 = -A$, then det (adj (A^n)) is
(A) 1, if n is even (B) –1, if n is odd
(C) 1 if n is odd (D) –1 if n is even

9. Which of the following when simplified reduces to unity?
(A) $\log_3 \log_{27} \log_4 64$ (B) $2\log_{18}\left(\sqrt{2}+\sqrt{8}\right)$
(C) $\log_2 \sqrt{10} + \log_2\left(\frac{2}{\sqrt{5}}\right)$ (D) $-\log_{\sqrt{2}-1}\left(\sqrt{2}+1\right)$

10. If $\int \frac{dx}{p^2 \sin^2 x + r^2 \cos^2 x} = \frac{1}{12}\tan^{-1}(3\tan x) + c$, then the value of p sin x + r cos x can be
(A) $\frac{6}{\sqrt{5}}$ (B) $\sqrt{5}$
(C) $6\sqrt{5}$ (D) –4

SECTION – C
(One Integer Value Correct Type)

This section contains **10 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. If $(x-a)(x-a^2) < 0$ has exactly one integral solution of x for $a \in I^+$, then a possible integral value of x is _____

2. If $|z_1 - 1| = 1$ and $|z_2 - 4i| = 1$, then the difference between maximum and minimum values of $|z_1 - z_2|$ is _____

Space for rough work

3. Two points A and B lie on the curve $y = x^2$ such that $\overrightarrow{OA}.\hat{i} = 1, \overrightarrow{OB}.\hat{j} = 4$, where O is origin and A and B lies in first and second quadrant respectively. Then $\overrightarrow{OA}.\overrightarrow{OB}$ equals _____

4. If $\log_{245} 175 = a, \log_{1715} 875 = b$, then the value of $\frac{1-ab}{a-b}$ is _____

5. A circular section is formed by a thread of fixed length then $\frac{\text{maximum area of section formed by thread}}{\text{maximum area of rectange formed by thread}}$ is equal to _____

6. For all positive real number x and y, such that x + y = 1, the maximum value of $36\left(x^4y + xy^4\right)$ is _____

7. The number of solutions (x, y, z) to the system of equations
$$x + 2y + 4z = 9$$
$$4yz + 2xz + xy = 13$$
$$xyz = 3$$
such that at least two of x, y, z are integers is _____

8. If number of points common between $||x| - |y|| = 1$ and $2a|x||y| + 1 = 2|x| + a|y|$ are 8 then value of 3a (where $a < 1$) is _____

9. The value of the expression
$\lim\limits_{y\to\infty}\left[(8y+3)(y+1)(y+2)(y+3)(2y+3)(4y+5)(2y+7)\right]^{1/7} - 2y$ is of the form $\frac{A}{B}$. Then the value $\left[\frac{A}{B}\right]$ (where [.] denotes the greatest integer function), is _____

10. If the number of the positive integral solutions of the equation $\left[\frac{x}{9}\right] = \left[\frac{x}{11}\right]$ is *n*, then $\frac{n}{3}$ is _____ (where [.] denotes the greatest integer function)

Space for rough work